24 October 2026 19:21

Week 1

Basic Recap of Python

Week 2

Selection sort, Insertion sort, Merge sort } Time complexity, Space complexity, Stability, In-place or Not

Time complexity (O(Big 'O') Upper bound
Θ Tight bound
Ω lower bound)

Default (List, Tuple, Dictionaries method's Time complexity)

	T.C (Average case)	S.C	Stability	In-place	
Selection sort	$O(n^2)$	O(1)	No	Yes	Not part of this course Execution memory, (Space complexity)
Insertion sort	$O(n^2)$	O(1)	Yes	Yes	
Merge sort (merge)	O(nlogn)	O(n)	Yes	No	
Quick sort (partition)	O(nlogn)	O(1) (Data storage)	No	Yes	

↳ Week 3

Linked list
↳ singly linked ○ → data .next
↳ doubly linked ←○→

Stack
↳ push (to add elements), pop (to remove elements)

l=[]
.append() (push) OR
.pop() (pop)

from collections import deque
q=deque()
↳ .append() (push)
↳ .pop() (pop)

Queue
↳ enqueue ("), dequeue (")

deque pronounced as 'deck'

l=[]
.pop(0) (dequeue) OR
.append() (enqueue)

from collections import deque
q=deque()
↳ .append() (enqueue)
↳ .popleft() (dequeue)

Hashing $h(x) = x \bmod m$
m is size of hashtable
Indexes are from (0 to m-1)
Collision resolution methods
↙ ↘
Open Addressing a.k.a close hashing ↙ ↘
Closed Addressing a.k.a open hashing ↳ linked list

a.k.a close hashing — Linear probing, Quadratic probing

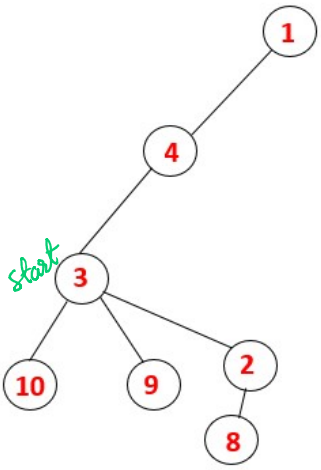
a.k.a open hashing → Linked list (seperate chaining)

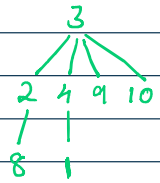
Week 4

Graphs

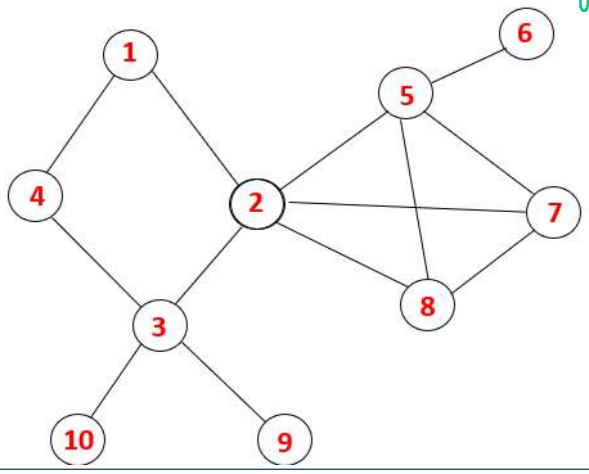
BFS, DFS

Implement DFS from (3)

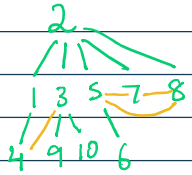




DFS from (2)

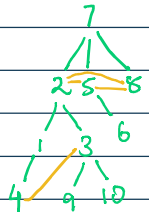


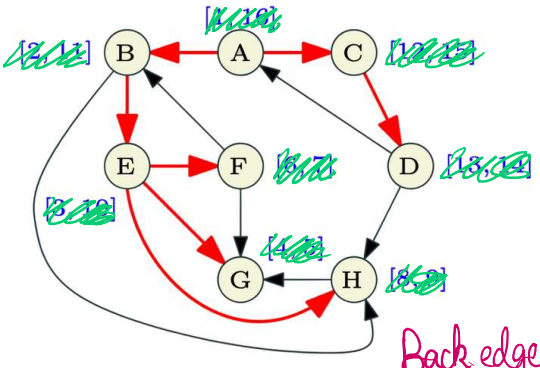
Search Tree

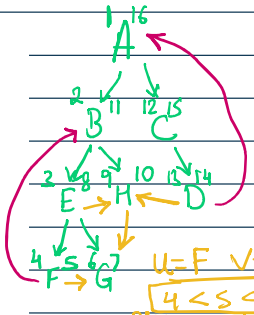


Cross edges:

Start = (7)







u=F v=G

4 < 5 < 6 < 7

pre(u) < post(u) < pre(v) < post(v)

Back edges ⊙ [pre(v), post(v)] contains [pre(u), post(u)]

Cross edges ⊙ [pre(u), post(u)] ★★ disjoint ★★ with [pre(v), post(v)]

pre(v) < pre(u) < post(u) < post(v)

v=B u=F

2 < 4 < 5 < 11

Forward edges ⊙

[pre(u), post(u)] contains [pre(v), post(v)]

Forward edges

[pre(u), post(u)] contains [pre(v), post(v)]

pre(u) < pre(v) < post(v) < post(u)

FE u contains v (u→v)

BE v contains u (u→v)

CE (Disjoint) (u→v)

^ ^ assigned ☆☆ with [pre(v), post(v)]

2 < 4 < 5 < 11

pre(v) < post(v) < pre(u) < post(u)

9 < 10 < 13 < 14

v=H u=D